

Promoting inclusion for disabled students in healthcare education: Using research-based theatre to enhance knowledge and empathy[☆]

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ABSTRACT

Aim: This study explores and compares the impact of in-person and online versions of a Research-based Theatre production named "Alone in the Ring," aimed at increasing knowledge and fostering positive attitudes towards persons with disabilities in healthcare professions. The study examines changes in attitudes towards disabled persons in healthcare professions, levels of engagement in the play and knowledge about the experiences and challenges of disabled individuals in healthcare professions. It also investigates possible actions to increase inclusion for disabled persons in healthcare education and practice.

Background: Disabled students face numerous barriers in healthcare education and practice, many of which are rooted in stigma and negative attitudes. Research-based Theatre promotes experiential learning to address these attitudes and stereotypes. The performance "Alone in the Ring" is based on a comprehensive study of the experiences of disabled students and clinicians in health professions, aiming to promote inclusion for students with disabilities in these fields.

Design: This study uses a pre-test and post-test design with two comparison groups.

Method: The study employed an explanatory sequential mixed method design to explore the impact of online versus in-person Research-based Theatre teaching in healthcare education. Participants were graduate health students (N = 198); 100 watched the in-person version and 98 watched the online version. Surveys were administered before and after the performance. Participants were also invited to share their experiences in semi-structured interviews and their responses were analyzed using thematic analysis.

Results: Results indicated that students generally felt more comfortable with disabled persons in healthcare professions in the post-survey compared to the pre-survey, with online participants reporting greater comfort and higher engagement than those in-person. Knowledge of disabled persons' experiences and challenges, as well as possible actions to increase belonging and inclusion, improved significantly for both groups from pre- to post-performance, with online participants showing a greater change. Qualitative data revealed that online theatre was perceived as authentic and beneficial for accessibility, engagement and emotional impact on students, though challenges included potential distractions and technological issues. Positive correlations were found between changes in knowledge, positive attitude and engagement, suggesting that increased engagement and a shift in attitudes can promote knowledge and awareness of the inclusion of disabled persons in healthcare professions.

Conclusion: These results support the use of online theatre-based academic teaching that experientially engages students, increases empathy and can promote more inclusive healthcare educational settings for disabled persons.

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1. Introduction

'Hi everyone, it's me, Samira. First, I want to apologize for not posting my vlog lately. My program is so overwhelming I literally don't have time to breathe... Remember when I shared with you that I got into grad school how thrilled I was? Thinking I was coming home – to a program where I could be myself.

I thought - after all, here people should understand disability, right?

I imagined how the faculty will say, "Welcome, you bring such a rich experience, you know the health system well, you learned empathy first hand."

But I'm so upset – you wouldn't believe what I'm hearing. The minute I said PTSD, it felt like a bomb dropped. I heard really difficult comments.

Someone actually told me: "You might want to find a different profession."'

(Samira, a first-year student in a health-education program; *Alone in the Ring: The Vloggers*, p. 8)

Healthcare education programs are known to be rigorous and demanding in nature and are more stressful for disabled students (Epstein et al., 2021; Mayer et al., 2024). Compared with non-disabled peers, healthcare students with disabilities face lower enrollment, poorer academic experiences and higher dropout rates (Bulk et al., 2017; Epstein et al., 2021). These findings have been attributed to multiple systemic barriers, including lack of accessibility and social stigma (Bulk et al., 2017; Epstein et al., 2021). As disabled individuals remain underrepresented in healthcare education and practice (Bulk et al., 2017, 2020), removing systemic barriers for these students constitutes a major step towards overcoming educational inequity (Jarus et al., 2022). Given that many of the systemic barriers faced by disabled healthcare students stem from ignorance or lack of compassion amongst peers, educators, or members of the public, a focus should remain on increasing education around this issue (Mirón Canelo et al., 2017; Wynants and Dennis, 2017).

2. Background

One way to develop an inclusive environment towards disabled people is educating healthcare students about the experiences of disabled people in healthcare education to create a more inclusive educational environment (Mayer et al., 2022). However, one of the challenges in addressing attitudes of healthcare students towards inclusion is finding ways to make the educational programs impactful.

One impactful way to address this challenge is Research-based Theatre (RbT). RbT has recently gained traction as an instructional tool for knowledge translation (Brown et al., 2018; Bird and Donelan, 2020). In using RbT as an arts-based learning approach, research findings are transformed into theatrical performance to engage audiences, stimulate the imagination and foster critical reflection (Cox and Belliveau, 2021). Further, RbT is particularly beneficial as a pedagogical approach because it allows audiences to experience stories more directly, catalyzing frank discussions in classes that would otherwise be difficult to elicit (Belliveau and Lea, 2016). As such, it serves as a powerful teaching tool in healthcare education, where there is a need to convey knowledge about human experience and enhance empathy among students who are training to become healthcare practitioners (Cox and Belliveau, 2021).

2.1. The application of RbT as a pedagogical approach to promote inclusion

The application of RbT as a pedagogical approach to promote equity,

diversity and inclusion has been well established (Belliveau and Lea, 2016; Freebody and Finneran, 2013). Multiple studies have demonstrated the efficacy of live performance in counteracting stigma towards vulnerable populations such as military veterans, grieving relatives of suicide victims and youth with depression (Belliveau et al., 2020; Hooyer et al., 2020; Sawyer and Earle, 2019; Silvén Hagström, 2020). In the healthcare context, RbT has been proven to support healthcare students and practitioners in developing a nuanced understanding of disabilities, chronic health conditions and other human experiences in the healthcare system (Cox et al., 2009; Cox and Belliveau, 2021; Kontos and Naglie, 2007; Schneider, 2017).

An RbT play called *Alone in the Ring* (AiTR) was created to translate the experiences of disabled healthcare students and clinicians who were interviewed multiple times over the course of a year, with a total of 126 interviews (described in Jarus et al., 2022). In 2020, the COVID-19 pandemic created a situation wherein presenting the in-person version of AiTR became untenable. The need for a quick solution to this problem, accentuated by a high demand for the performance coming from universities in other geographic areas, pushed the team towards creating a new, online version. Implementing this online RbT performance became a meaningful way to reach healthcare education students and speak about inclusion in healthcare professions.

2.2. The current study

While technology has been widely explored in theatre education contexts (Vickers, 2020), the online delivery of RbT is relatively uncharted territory (Cook et al., 2020). This study explores how AiTR, which was previously part of an in-person academic orientation program for healthcare students, was transformed into an online program. The study specifically compares the impact and efficacy of the in-person versus the online version on attitudes of students towards disabled students and clinicians with disabilities.

The online version, *Alone in the Ring: The Vloggers*, was performed via Zoom. The subtitle highlights a significant change in the version of the play as previous monologues and dialogue from characters were formatted into a series of pre-recorded vlogs streamed at the beginning of each performance. In turn the climactic scene wherein the in-person version the characters were in a boxing ring representing the struggles of stigma faced by disabled students and clinicians, the action was re-configured into a live virtual panel with discussions taking place between the characters.

The in-person and the online performances were similar with regards to the following elements: a) length of performance; b) actors involved; c) themes and narratives conveyed; d) characters; e) the final call for audience members to take specific actions to support creating more inclusive workspaces; and f) additional information imparted post-performance. The in-person version of AiTR was performed for groups of healthcare students in a specific year and the online version for the next cohort of students in the same programs one year later, during the pandemic.

To our knowledge, this is the first study comparing the efficacy of in-person and online RbT performances as an academic instructional tool to increase inclusion in healthcare education. Therefore, this study examined two exploratory questions: (a) Was there a difference between the level of engagement students reported while watching the in-person versus online version of AiTR? (b) Was there a difference in the changes of knowledge or attitudes towards disabled health students and clinicians reported for students who watched the in-person versus the online version of AiTR?

3. Method

3.1. Study design

This study used an explanatory sequential mixed method approach

to evaluate the impact of the AiTR performances on knowledge and attitudes regarding disability equity in healthcare professions. The quantitative component involved survey measures administered at two time-points, one immediately before and one immediately after the performance. The qualitative component involved follow-up interviews and focus groups to complement quantitative findings.

3.2. Recruitment and participants

Participants in both versions were recruited through the university's healthcare professional programs. All students were invited to watch the play as part of a first week orientation for their education programs. AiTR was presented in English as were the pre- and post- quantitative and qualitative measures. In total, there were 100 participants from two in-person performances that were conducted in August and September 2019 and 98 from two online performances that were conducted on September 2020. No differences were found between the groups in gender ratios ($\chi^2(2) = 2.210, p = .33$), age ($\chi^2(4) = 3.154, p = .53$) or disability ($\chi^2(2) = .885, p = .71$). Eleven online participants consented to an interview about their online experiences. In person participants were not interviewed in this study, as the focus of the study was using RbT as an online pedagogy. Interview participants are described in Table 1.

3.3. Study procedure

The project was approved by the University of British Columbia's Behavioral Research Ethics Board. A week before the performances, registered students were sent a letter detailing the purpose of the research, implied consent and options for voluntary participation in the discussion, surveys and interviews. The survey also included an option to indicate an interest to be contacted for an interview. All students could choose to join the performance without participating in the study.

Both performances lasted about 35 minutes. The in-person performance included monologues, dialogues and a boxing ring scene. The online performance included a series of vlog-style clips and a live panel. Both performances shared characters' struggles and triumphs in navigating their role as a healthcare student or professional with a disability. At the end of both versions, there was a discussion session involving the research team and actors — several of the actors were researchers themselves. Three out of the four actors are disabled persons. Ethically it is important that the experiences of disabled persons are portrayed directly by individuals with lived experiences (Michalovich et al., 2023).

Table 1
Participants' demographic data by group.

	In-person AiTR (n = 100)	Online AiTR (n = 98)	The interview participants (n = 11, 11.22 % Online AiTR)
Age			
18–25	58 (58 %)	72 (73.5 %)	9 (81.8 %)
26–30	23 (23 %)	20 (20.4 %)	2 (18.1 %)
31–40	8 (8 %)	5 (5.1 %)	
41–50	1 (1 %)	1 (1 %)	
51 and up	1 (1 %)		
Prefers not to answer	9 (9 %)		
Gender			
Female	70 (70 %)	78 (79.6 %)	10 (90.9 %)
Male	26 (26 %)	20 (20.4 %)	1 (9.09 %)
Non-binary	1 (1 %)		
Prefers not to answer	3 (3 %)		
Having a disability			
Yes	15 (15 %)	16 (16.3 %)	2 (18.18 %)
No	81 (81 %)	80 (81.6 %)	5 (45.45 %)
Prefers not to answer or unsure	4 (4 %)	2 (2 %)	4 (36.36 %)

The team included trained mental health professionals who were available to support any audience members triggered by the content. A total of four performances were offered: two in-person and two online.

Immediately before the performance, participants completed an online “pre” survey containing a demographic questionnaire, a knowledge measure and a baseline attitude measure. Another survey was completed right after the performance but before the discussion. The “post” survey contained the same attitude measure and had additional questions about knowledge on equity in healthcare professions and level of engagement during the performance. In the in-person version, the questions were presented on screen and read out loud and participants were invited to use clickers to indicate their choices. In the online version, a Qualtrics survey was shared via email before the performance and through the chat after the performance.

Healthcare students who consented to be contacted were emailed a request to participate in a follow-up interview or focus group. After obtaining written consent, semi-structured interviews were conducted to discuss students' experiences. Another study on this project explores the audience's experiences with the in-person performances (Jarus et al., 2022). All interviews were conducted over Zoom, audio recorded and transcribed verbatim. Interview participants received a \$30 honorarium.

4. Tools

4.1. Attitude scale

The British Social Attitudes Survey (BSAS), used by the National Centre of Social Research since 1983, was modified to address participants' perceptions of disabled healthcare students and clinicians (Staniland, 2011). The original survey included a series of questions regarding public attitudes towards disabled individuals. These included questions that asked ‘how comfortable’ respondents would feel interacting with disabled people in different situations. We adapted this format to create five short scenarios depicting disabled students and clinicians including mental, physical and sensory disabilities. After each scenario depicting a short interaction of a student or clinician telling their client or colleagues that they have a specific disability, the participants responded to the question, ‘If you were the client/colleague/supervisor, how would you feel in this situation?’ on a 4-point Likert scale ranging from ‘Very uncomfortable’ (1) to ‘Very comfortable’ (4). The attitude scale was administered both prior to (T1) and after (T2) the performance. The scenarios were presented by the actors for the in-person version and were filmed and embedded as short videos in the Qualtrics survey in the online version. Cronbach Alphas were .63 for T1 attitude items and .80 for the T2 attitude items.

4.2. Knowledge measure

Prior to the performance (T1), one generic question was asked to measure the extent to which participants were familiar with the experiences of disabled individuals in healthcare professions. This was done using a 5-point Likert scale, ranging from ‘I don't know anything - 0 % from what there is to know on the topic’ (1) to ‘I know very much - About 100 % from what there is to know on the topic’ (5). After the performance (T2), four questions to assesses the change in knowledge as perceived by the participants with the preamble: ‘Compared to what you knew before, how much did you learn about...?’ The questions focused on knowledge about (a) barriers, (b) stigma, (c) disclosure and (d) actions of allyship. Using a 5-point Likert scale and similar to the T1 question, the answers ranged from ‘None - 0 % compared to what I knew’ (1) to ‘Substantially - 100 % more than what I knew’ (5). We also asked one question about change in motivation to act where the answers ranged from ‘Less motivated than before’ (1) to ‘Substantially - 100 % more motivated’ (5). Cronbach alpha was .78 for the T2 knowledge items.

4.3. Emotional engagement measure

To measure the levels of engagement, the Emotional Engagement subscale from the Narrative Engagement Scale was used (Busselle and Bilandzic, 2009). This subscale includes three items that measure emotions that audiences may have with respect to characters, either feeling the characters' emotions (empathy), or feeling for them (sympathy) (e.g.: 'The story affected me emotionally'). Participants were asked to rate their level of agreement on a Likert scale ranging from 1 (strongly disagree) to 4 (strongly agree). Cronbach alpha was .56 for the engagement scale.

4.4. Interview guide

We used an interview guide to ask the participants about their experiences over Zoom. Questions included items such as: 'Have you ever seen a performance online? Have you heard before about research-based theatre?', 'What was your experience watching the performance online?', 'What did you feel were any advantages or disadvantages of watching it online?', 'Did anything about the online performance surprise you?'

4.5. Data analysis

Quantitative data from the surveys were analyzed with SPSS version 26 (IBM Corporation). To assess interactions between performances (in-person and online), time (T1 and T2) and attitude subscales scores, a repeated measures ANOVA was performed with within-subject factors 'time' (2) by 'scenario' (5) and between-subjects factor 'performance' (2). As Mauchly's Test of Sphericity was significant on the attitudes scale, sphericity could not be assumed and thus the Greenhouse-Geisser correction was used. Repeated ANOVA tests were also employed to observe the effect of performance on knowledge and engagement subscales.

Qualitative data from interviews were coded through thematic analysis to identify major themes and subthemes. All the names of the interviewed students are pseudonyms chosen by the participants or assigned by the research team, while maintaining the cultural orientations. Transcripts were analyzed by two members of the research team to develop a coding framework: generating initial codes, searching for themes, reviewing themes, defining and naming the themes and producing the report (Braun and Clarke, 2006). Themes that emerged from the qualitative data helped to interpret quantitative results (Klassen et al., 2012).

5. Results

As the focus of this paper is on the differences between the in-person and Zoom performances, we only elaborate on main effects or interactions that include the type of performance variable.

a. Effect on attitudes

The results of the repeated measure ANOVA indicated that there was a significant main effect of scenario type, with a large effect size (see Table 2). A significant main effect was observed for Time with a large effect size. Regardless of the version, participants were significantly more comfortable with the scenarios in the post-survey than in the pre-survey. There was a significant main effect for Performance for the attitude scale, with a large effect size. Participants from the online performance reported greater comfort with the scenarios than those from the in-person performance. The three-way interaction between scenario by Time and by Performance was not found to be significant. However, the interaction between the in-person and online versions and Scenario was significant. Post hoc analysis revealed that while overall participants in the Zoom performance reported being more comfortable with all scenarios than

Table 2

Results of ANOVA analysis.

a. Results of the ANOVA of Scenario by Time by Performance				
	<i>F</i>	<i>df</i>	<i>p</i>	η^2 ^a
Scenario	53.558	4	.000	.215
Time	121.312	1	.000	.382
Performance	39.992	1	.000	.169
Time * Scenario	19.272	4	.000	.102
Time * Performance	.068	1	.795	.000
Performance * Scenario	6.223	4	.000	.031
Time * Performance * Scenario	1.638	4	.163	.008
b. Results of the ANOVA of engagement by performance				
	<i>F</i>	<i>df</i>	<i>p</i>	η^2 ^a
Scales	6.106	1	.014	.030
Performance	4.973	1	.027	.025
Performance * Scales	.353	1	.553	.002
c. Results of the ANOVA of change in knowledge and performance				
	<i>F</i>	<i>df</i>	<i>p</i>	η^2 ^a
Time	83.347	1	.000	.298
Performance	6.238	1	.013	.031
Performance * Time	6.561	1	.011	.032
d. Results of the ANOVA of post knowledge items by performance				
	<i>F</i>	<i>df</i>	<i>p</i>	η^2 ^a
Type of knowledge	39.758	4	.000	.169
Performance	15.601	1	.000	.074
Performance * Type of knowledge	2.626	4	.034	.013

p* < .05; *p* < .01; ****p* < .001

^a Eta sq.: small es.01-.058; medium es.059-.137; large es > .137

those at the in-person performance, those differences were significant only for two of the scenarios: the blind counsellor and the physiotherapist with a cane. Table 2 and Fig. 1 present the interaction between Performance and scenario type on the attitude level (i. e. feeling comfortable in the scenario of interacting with a disabled student/clinician).

b. Effect on engagement

Engagement scores were significantly higher, with a small effect size, in the online performance group compared with in-person (see Table 2, Fig. 2). Additionally, a significant, small main effect was found for the engagement scale; online participants reported being more empathetic towards the characters and their experiences rather than being emotionally affected by the story itself. There was no significant interaction between performance and scale.

c. Effect on knowledge

The results of the repeated measure ANOVA indicated that there was a significant main effect for Time and Performance type, with large and small effect sizes (see Table 2). Based on the main effect, both groups reported improved knowledge from pre to post performance. However, the post hoc results of the significant interaction between performance and knowledge (with small effect size) indicated that participants in the online performance group reported a greater change compared with the in-person group (see Table 2).

Finally, the results of the repeated measure ANOVA analyzing the difference between the knowledge scales in the post-performance and between types of performances indicated that there was a significant main effect for scale and performance type, with large and medium effect size correspondingly (see Table 2). The significant interaction, with a small effect size, between knowledge scale and performance type superseded the main effects. Post hoc analysis indicates that participants in the online performance reported higher knowledge scores in all subscales, the differences were only significant for the stigma and motivation subscales (see Fig. 3 and Table 2).

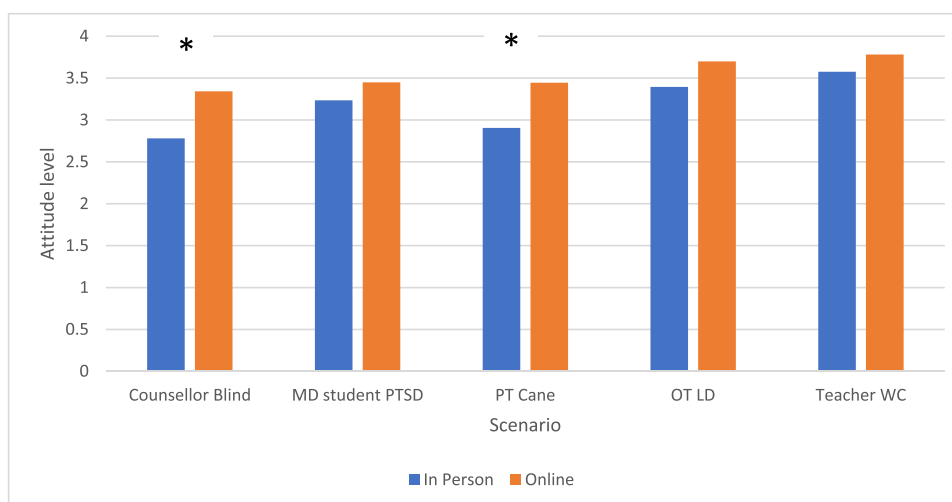


Fig. 1. The interaction between the scenario type and performance type on participants' attitude level.

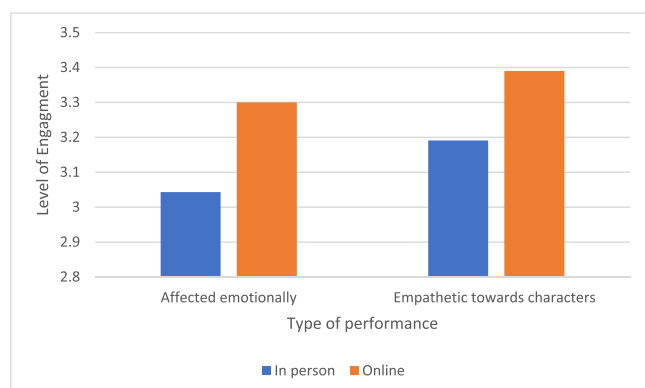


Fig. 2. Type of performance and type of engagement relations with level of engagement.

5.1. Correlations between change in knowledge, change in attitude and engagement scores

Pearson's correlation tests revealed significant moderate, positive associations between reported change in knowledge with reported change in attitude and engagement for the in-person and online performance groups (with similar strength of association). This suggests that engagement in the learning material and a shift in attitude can promote increased knowledge and awareness (see [supplemental material Table 3](#)). No correlations were found between engagement and change in attitude.

5.2. Qualitative results

Four major themes emerged from the data: (a) experiences and expectations of online theatre teaching, (b) benefits of online theatre teaching, (c) disadvantages of online theatre and (d) feedback for future online performances.

5.3. Expectations and experiences of online theatre

All participants described AiTR to be their first experience of online theatre. Participants were generally impressed with the authenticity of the experience. Participants described the experience as genuine and realistic. Muhammad said: 'This one did feel very genuine, like the characters' perspectives and experiences really got me invested.' Many

were emotionally engaged and felt compelled to help characters that were bullied within the chat. Kelly said: 'I think I was surprised by how pressured I felt to kind of jump in and do something about the chat!'

All participants described an immediate impact of the performance, discussing it with peers in class and reflecting on the unique nature of their learning experience. A student with a disability, Dhriti, described the impact of the performance in allowing her to feel more comfortable in her program by promoting a more welcoming environment:

I am someone that I classify as having a disability or a chronic health condition. So I definitely could relate to a lot of the experiences. And I thought, wow, this makes me feel more comfortable knowing that, like, my classmates and colleagues would be able to see the perspectives and sort of have that more insight.

5.4. Benefits of having online theatre performance

Participants had a very positive impression of the AiTR online performance, citing multiple advantages of having the play delivered online. The following sub-themes demonstrate the strengths of the AiTR online performance.

a. Accessibility and Distribution

Perhaps the most widely acknowledged advantages to online theatre are greater accessibility and convenience as well as potential for widespread distribution. Participants appreciated the convenience of participating from the comfort of their homes and recognized that, for those with limitations for mobility or transportation, eliminating the need to physically attend performances could make a significant impact on accessibility; Kirra stated that 'with obviously COVID, it kept everyone safe.' Lyn described how the Zoom performance was a refreshing experience compared with other online learning: 'I thought that it was just perfect. I haven't really seen that much creativity in regard to Zoom before.'

Another participant discussed psychological safety when watching the performance from home. Kelly stated: 'I definitely see there being benefits to people kind of engaging with some of these tougher topics from an environment that they're more comfortable with. Like for me, I was at home and I'm watching some things that could be kind of difficult or triggering for some people, so I thought that was good to kind of be able to be in a safe space.'

b. Manipulation of theatrical elements and opportunities for engagement and discussion.

Some participants described different opportunities within Zoom for presenting the content. Kim said: 'The advantage of being online

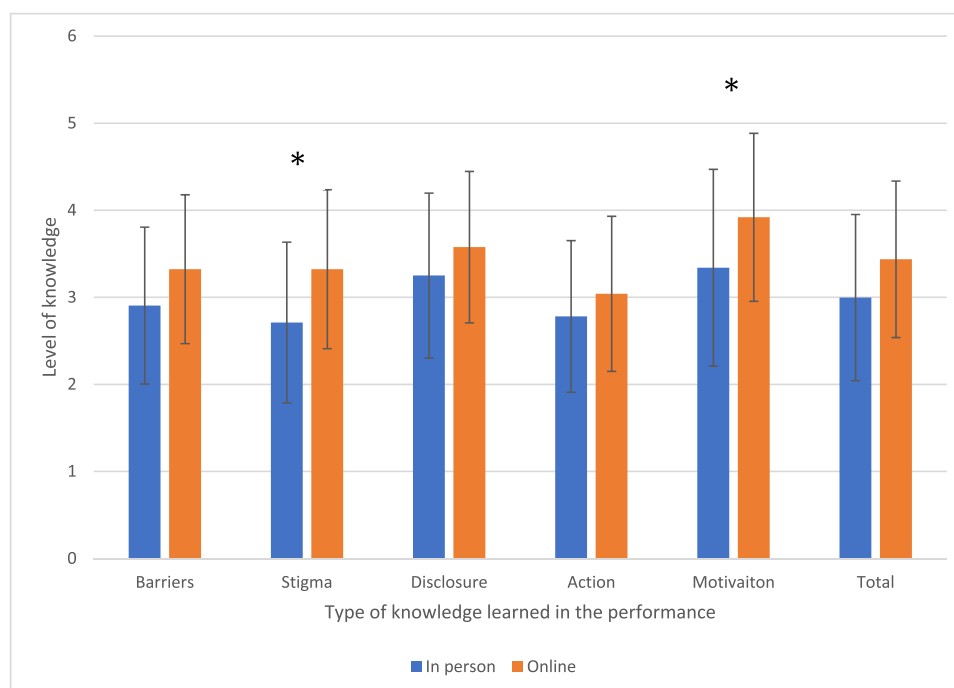


Fig. 3. Type of knowledge and performance impact on level of knowledge.

is that you can have that blend of being recorded and, like, switching between screens and having it look like you're just watching it on YouTube, which I feel like some of the presenters did that so good. It was, like, more immersive in that sense.'

Some participants also noted the emotional appeal of the online delivery. Ally referred to a scene filmed in a closet featuring a nurse experiencing an anxiety attack at work, 'you could almost feel like you were also in that closed space [...] almost that "alone in the ring" feeling.' The online format thus appears to facilitate manipulation of environment and atmosphere for emotional impact, creating a sense of proximity to the characters: 'I feel like the selfie camera [...] gives you that intimate moment with the character' (Mau). Similarly, Kim expressed:

I was surprised, just watching, like, a recording. And because it was so raw, it wasn't produced; it kind of looked like it was just him filming on his phone, in a closet. It felt very real... I felt quite emotional watching that because he was going through an anxiety attack, it just felt really personal, so I felt really connected watching that.

Another benefit of the online format is the possibility to film scenes in different settings or using the camera to convey a sensation like, for example, obscuring it to create an experience of blindness. Kim said: 'I liked that... the one that was blurry, I like that she kind of put us in her shoes.'

Additionally, the ability to use the chat function was advantageous. It encouraged more discussion between students, particularly among those who are 'a little bit more reserved to speak up or say something [in-person]' (Ally). This invitation for the students to interact with the characters added to the play's educational value by promoting engagement: 'I really enjoyed that there was an opportunity to engage at the end [...] because I personally have a really hard time paying attention to online classes' (Kirra). Lyn talked about the use of the chat box as one of the most surprising and engaging components in the performance: 'I think I was just surprised that it became live, like, that I wasn't expecting. And at first, I was kind of like, what's going on? And who are these people in the chat box? Why are people saying this? And then I was like, Oh, this is part

of the performance... there was an element of surprise there, which kept it interesting!'. Participants were keen to see more opportunities for audience engagement and interaction in future online performances and teaching.

c. *The experiential dimension in comparison to traditional online academic teaching*

All participants mentioned that learning online, which was common during the pandemic, was 'dry' for them and the novelty of the performance sparked their curiosity and interest in inclusion issues. Participants also described how the theatre component increased empathy and perspective taking. For example, Muhammad said: 'I really think it got you to feel the empathy for the characters... And I think you just made you feel like another level of empathy because you really could view them as real people.' And For example, Leila said: 'It brought to life the experiences of [disabled] people in the healthcare system and their emotions and their feelings... we maybe wouldn't have picked up on if we just read about a story or, like, a statistic or something like that.'

d. *Disadvantages of online theatre performances*

While participants were impressed by the effectiveness of the online delivery, they also highlighted possible disadvantages. First, the general impression that online activities are 'harder to engage in' was echoed by nearly all participants. This was attributed to more potential distractions, technological problems and emotional distance: 'You don't get the interpersonal connections that you may feel from watching characters in-person.' (Danny). Especially given the restrictions on social gatherings during the pandemic, it is natural for individuals to 'miss[ed] the in-person interaction' (Ally).

Accessibility emerged as another concern owing to the necessity of the individual having access to and familiarity with technology. Lack of high-speed internet connection and personal difficulties using technology were identified as major barriers for efficient learning. Interestingly, the delivery of this play over Zoom, in parallel with its focus around inclusion, prompted some participants to reflect on accessibility issues surrounding online learning, such as the absence of 'closed captioning for most lectures, or different formats of learning,' (Danny).

6. Discussion

The purpose of this study was to explore the experiences of healthcare education students in an online performance as an instructional tool. Survey data demonstrated a positive change in attitude and knowledge for both in-person and online groups. However, participants from the online performance reported greater comfort with the scenarios than those attending the in-person performance, especially with two scenarios involving disabled clinicians providing a service. In addition, the online participants reported greater engagement in the narrative and a higher increase in knowledge in comparison with in-person participants. The relationships between engagement levels and knowledge change may suggest that emotional involvement was related to increased learning (Tyng et al., 2017). While there is limited literature comparing these dimensions across in-person and online settings, there have been numerous studies demonstrating the efficacy of online formats (Hill et al., 2020; Leslie et al., 2015; Wynants and Dennis, 2017). Similarly, in this study, online participants not only had a change in attitude but also a better understanding of the topic at hand after the performance. As discussed, the dramatic effect of the online environment helped foster an emotional impact and increased empathy for the students.

Higher engagement scores amongst online participants challenges the general perception that students are less engaged with online learning formats (Butnaru et al., 2021). The novelty of the play's online format and the interactive discussion helped to promote interest and engagement. Similarly, student-student and student-instructor interactions are significant contributors to student satisfaction with online learning (Andrew et al., 2015; Croxton, 2014). The vlogging format received considerable praise for its creativity and authenticity. The role of vlogging in present society has extended beyond that of entertainment to include social advocacy and education (Raby et al., 2018). As such, the online *AiTR* performance taps into a familiar structure of communication that many students can relate to. This study demonstrates the potential for online RbT as a pedagogical tool for knowledge exchange that can promote inclusion and equity for disabled people in healthcare programs.

6.1. Limitations and implications for healthcare academic teaching

There are several limitations to address in this study. First, the in-person and the online performances were conducted with different cohorts and at different times which did not allow for randomized assignment. To control for any demographic factors, we compared variables and found no differences between the groups. Second, when we compared changes in knowledge, we relied on participant perceptions and assessment. In future studies, other means of knowledge assessment may be used in addition to more delayed assessment following the play. Future studies can also benefit from exploring the qualitative differences among in-person attendees in comparison to online attendees by interviewing in-person attendees. Additionally, the study was conducted in one university, in a large city, with only two specific health programs. Future studies should be implemented in other cultural, geographical and academic settings. Lastly, we could not measure any specific actions taken by participants. Future studies should consider investigating the efficacy of online RbT — and RbT in general — on action-taking. These findings may help inform whether dissemination of RbT performances is an effective strategy for fighting social stigma and making healthcare educational programs and workplaces more inclusive for disabled service users and providers.

In conclusion, our findings demonstrate the efficacy of an RbT performance at changing attitudes and increasing knowledge in the area of equity and inclusion. Since students and faculty are becoming increasingly proficient with online teaching and learning, new opportunities may become available to provide accessible programs to promote inclusion in education. Although creating a virtual performance is not an

easy endeavor, once created, it has the potential to reach broad audiences in remote areas and the RbT methodology enables the production of small, unfunded projects. RbT is an approach aimed to be useful and relevant for instructors and faculty members who, working with the support of artists, are able to create small productions that can be shared in different classrooms (Cox and Belliveau, 2021). Since healthcare students are very much used to engaging with video content, using this mode of delivery in academic teaching is promising and exciting, especially to fight disability stigma.

The conclusions of this study highlight several important implications for healthcare education in practice. Using engaging strategies, such as RbT, can enhance an inclusive climate for disabled healthcare students. Online formats could be particularly effective in fostering emotional involvement and empathy, which can lead to better learning outcomes. Educators could integrate these engaging methods alongside traditional approaches to create a more inclusive and accessible learning environment. Additionally, incorporating interactive and familiar structures that students relate to into healthcare education can further promote inclusivity and equity, making healthcare educational programs more welcoming for disabled service users and providers.

CRedit authorship contribution statement

Tal Jarus: Writing – review & editing, Validation, Supervision, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Kathy Xie:** Writing – review & editing, Investigation, Formal analysis, Data curation. **George Belliveau:** Writing – review & editing, Supervision, Resources, Methodology, Investigation, Funding acquisition, Conceptualization. **Christina Cook:** Writing – review & editing, Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Laura Yvonne Bulk:** Writing – review & editing, Validation, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Laen Dov hershler:** Writing – review & editing, Validation, Project administration, Methodology, Funding acquisition, Data curation, Conceptualization. **Yael Mayer:** Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of Competing Interest

The authors have no conflict of interest to disclose.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.nepr.2024.104085.

References

- Belliveau, G., Cox, S., Nichols, J., Lea, G.W., Cook, C., 2020. Examining the ethics of research-based theatre through Contact! Unload. In: *The Routledge companion to applied performance*. Routledge., pp. 141–148.
- Belliveau, G., W. Lea, G., 2016. *Research-Based Theatre: An Artistic Methodology*. University of Chicago Press, Chicago, IL.
- Bird, J., Donelan, K., 2020. An interactive ethnographic performance: using research-based theatre and drama pedagogy to enhance meaning making. *Res. Drama Educ. J. Appl. Theatre Perform.* 25 (2), 218–235. <https://doi.org/10.1080/13569783.2020.1729717>.
- Braun, V., Clarke, V., 2006. Using thematic analysis in psychology. *Qual. Res. Psychol.* 3 (2), 77–101. <https://doi.org/10.1191/1478088706qp0630a>.
- Brown, A., Ramsay, N., Milo, M., Moore, M., Hossain, R., 2018. How research-based theatre is a solution for community engagement and advocacy at regional medical campuses: The Health and Equity through Advocacy, Research and Theatre (HEART) program. *Can. Med. Educ. J.* 9 (1), e6.
- Bulk, L., Easterbrook, A., Roberts, E., Groening, M., Murphy, S., Lee, M., Ghanouni, P., Gagnon, J., Jarus, T., 2017. We are not anything alike': Marginalization of health

- professionals with disabilities. *Disability & Society* 32 (5), 615–634. <https://doi.org/10.1080/09687599.2017.1308247>.
- Bulk, L.Y., Tikhonova, J., Gagnon, J.M., Battalova, A., Mayer, Y., Krupa, T., Lee, M., Nimmon, L., Jarus, T., 2020. Disabled healthcare professionals' diverse, embodied, and socially embedded experiences. *Advances in Health Sciences Education* 25, 111–129.
- Butnaru, G.I., Niță, V., Anichiti, A., Brinză, G., 2021. The effectiveness of online education during COVID-19 pandemic—a comparative analysis between the perceptions of academic students and high school students from Romania. *Sustainability* 13 (9), 5311. <https://doi.org/10.3390/su13095311>.
- Cook, C., Shigematsu, T., Belliveau, G., 2020. Teaching Research-Based Theatre Online: A Narrative of Practice. *ArtsPraxis* 7 (2), 56–69.
- Cox, Susan, George, Belliveau, 2021. Health Theatre: Embodying Research." In *Popularizing Scholarly Research*. In: *Research Methods and Practices*, 202–29. Oxford University Press.
- Cox, S.M., Kazubowski-Houston, M., Niskier, J., 2009. Genetics on stage: public engagement in health policy development on preimplantation genetic diagnosis. *Soc. Sci. Med.* 68 (8), 1472–1480. <https://doi.org/10.1016/j.socscimed.2009.01.044>.
- Croxton, R.A., 2014. The role of interactivity in student satisfaction and persistence in online learning. *J. Online Learn. Teach.* 10 (2), 314.
- Epstein, I., Khanlou, N., Ermel, R.E., Sher, M., Simmonds, K.K., Balaquiao, L., Chang, K. Y., 2021. Students who identify with a disability and instructors' experiences in nursing practice: a scoping review. *Int. J. Ment. Health Addict.* 19, 91–118. <https://doi.org/10.1007/s11469-019-00129-7>.
- Freebody, K., & Finneran, M. (2013). Drama and social justice: Power, participation and possibility. *How Drama Activates Learning: Contemporary Research and Practice*, 47.
- Hill, B., Tulloch, M., Mlcek, S., Lewis, M., 2020. The 'within' journey: assessment of the online Indigenous Australian cultural competence training programme at Charles Sturt University. *Aust. J. Indig. Educ.* 49 (1), 14–22. <https://doi.org/10.1017/jie.2018.12>.
- Hooyer, K., Ruffalo, L., Franco, Z., 2020. Tracings of trauma: Engaging learners and challenging veteran stigma through collaborative research-based theater. *J. High. Educ. Outreach Engagem.* 24 (1), 127–142.
- Jarus, T., Krupa, T., Mayer, Y., Battalova, A., Bulk, L.Y., Lee, M., Nimmon, L., Earlene, R., 2022. Negotiating legitimacy and belonging: Disabled students' and practitioners' experience. *Medical Education* 57 (6), 535–547. <https://doi.org/10.1111/medu.15002>.
- Jarus, T., Mayer, Y., Gross, E., Cook, C., Bulk, L.I., Hershler, L.A.D., Nicholes, J., Zaman, S., Belliveau, G., 2022. Bringing disability experiences front stage: Research-based theatre as a teaching approach to promote inclusive health education. *Nurse Education Today* 115, 105408. <https://doi.org/10.1016/j.nedt.2022.105408>.
- Klassen, A.C., Creswell, J., Plano Clark, V.L., Smith, K.C., Meissner, H.I., 2012. Best practices in mixed methods for quality of life research. *Qual. Life Res.* 21, 377–380. <https://doi.org/10.1007/s11136-012-0122-x>.
- Kontos, P.C., Naglie, G., 2007. Expressions of personhood in Alzheimer's disease: An evaluation of research-based theatre as a pedagogical tool. *Qual. Health Res.* 17 (6), 799–811. <https://doi.org/10.1177/1049732307302838>.
- Leslie, A., Beverley, E., Sian, M.P., 2015. Enhancing the online learning experience using virtual interactive classrooms. *Aust. J. Adv. Nurs.* 32 (4), 22–31. (<https://search.informit.org/doi/10.3316/informit.430537564307700>).
- Mayer, Y., Shalev, M., Nimmon, L., Krupa, T., Bulk, L.Y., Battalova, A., Lee, M., Jarus, T., 2022. The impact of social support interactions on the psychological safety of students and practitioners with disabilities in health professions. *Advances in Health and Science Education* 28 (2), 477–497. <https://doi.org/10.1007/s10459-022-10169-9>.
- Mayer, Y., Shiffman, N., Etgar, S., Lurie, I., Jarus, T., 2024. The Mental Health Toll: Medical Trainees Living with Disabilities during the COVID-19 Pandemic. *Health & Social Care in the Community*.
- Michalovich, A., Mayer, Y., Hershler, L.A.D., Bulk, L.Y., Cook, C., Graf, H., Jarus, T., 2023. Through a glass brightly: Generative ethical tensions in Research-based Theatre. *Qualitative Inquiry* 29 (2), 267–276.
- Mirón Canelo, J.A., Iglesias de Sena, H., Alonso Sardón, M., Sánchez Fuentes, S., Fernández Lázaro, C.I., Chamorro Fernández, A.J., Santacruz Ruis, S., Hernández Galilea, E., José Ángel Santos, Sánchez, Fernández Martin, C., Lorenzo-Gómez, M.F., 2017. Inclusive education in medicine: awareness health care for persons with disabilities. *Educ. Med.* <https://doi.org/10.1016/J.EDUMED.2017.07.013>.
- Sawyer, J., Earle, S.J., 2019. Applied theatre as participatory action research: empowering youth, reframing depression. *Int. J. Innov. Creat. Change* 4 (4), 83–97.
- Schneider, J., 2017. Evaluation of the impact on audiences of inside out of mind, research-based theatre for dementia carers. *Arts Health* 9 (3), 238–250. <https://doi.org/10.1080/17533015.2016.1251475>.
- Silvén Hagström, A., 2020. Based theater and "stigmatized trauma": the case of suicide bereavement. *Front. Psychol.* 11, 1129. <https://doi.org/10.3389/fpsyg.2020.01129>.
- Tyng, C.M., Amin, H.U., Saad, M.N., Malik, A.S., 2017. The influences of emotion on learning and memory. *Front. Psychol.* 8, 235933 <https://doi.org/10.3389/fpsyg.2017.01454>.
- Vickers, S., 2020. Online theatre voice pedagogy: a literature review. *Voice Speech Rev.* 14 (3), 254–268. <https://doi.org/10.1080/23268263.2020.1777691>.
- Wynants, S.A., Dennis, J.M., 2017. Embracing diversity and accessibility: a mixed methods study of the impact of an online disability awareness program. *J. Postsecond. Educ. Disabil.* 30 (1), 33–48.